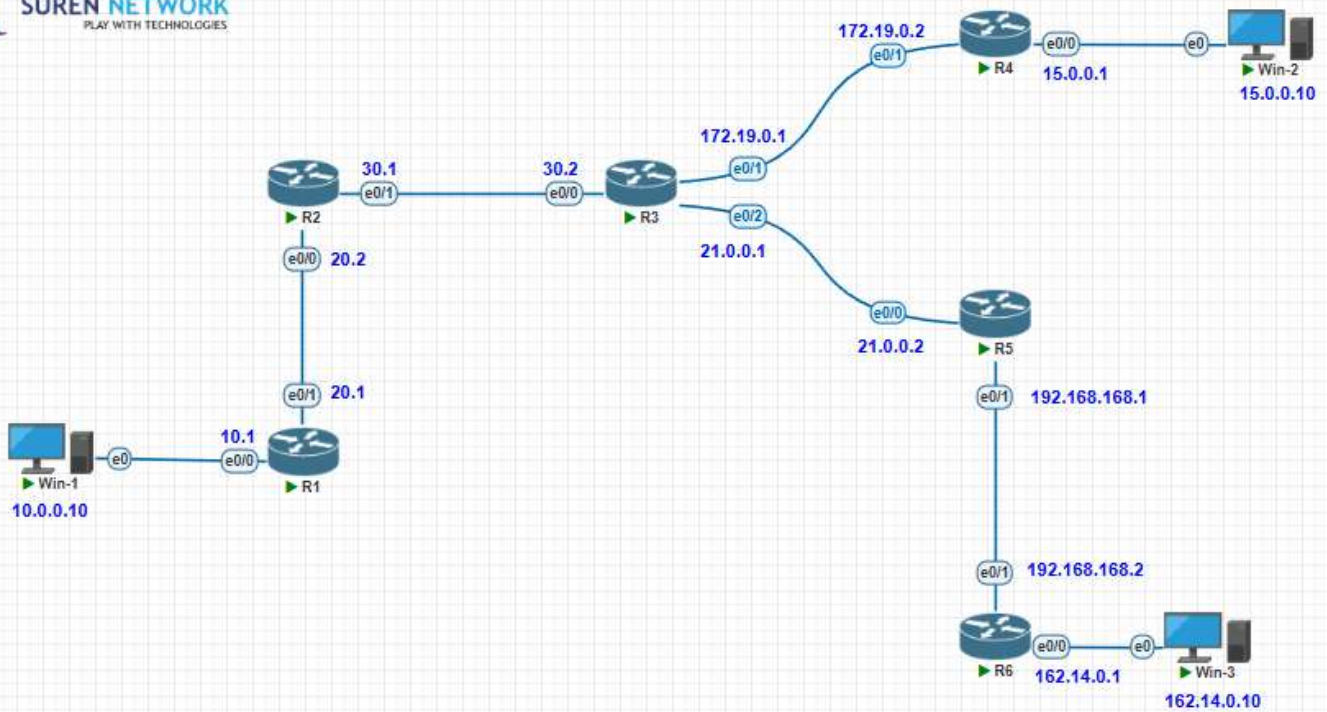
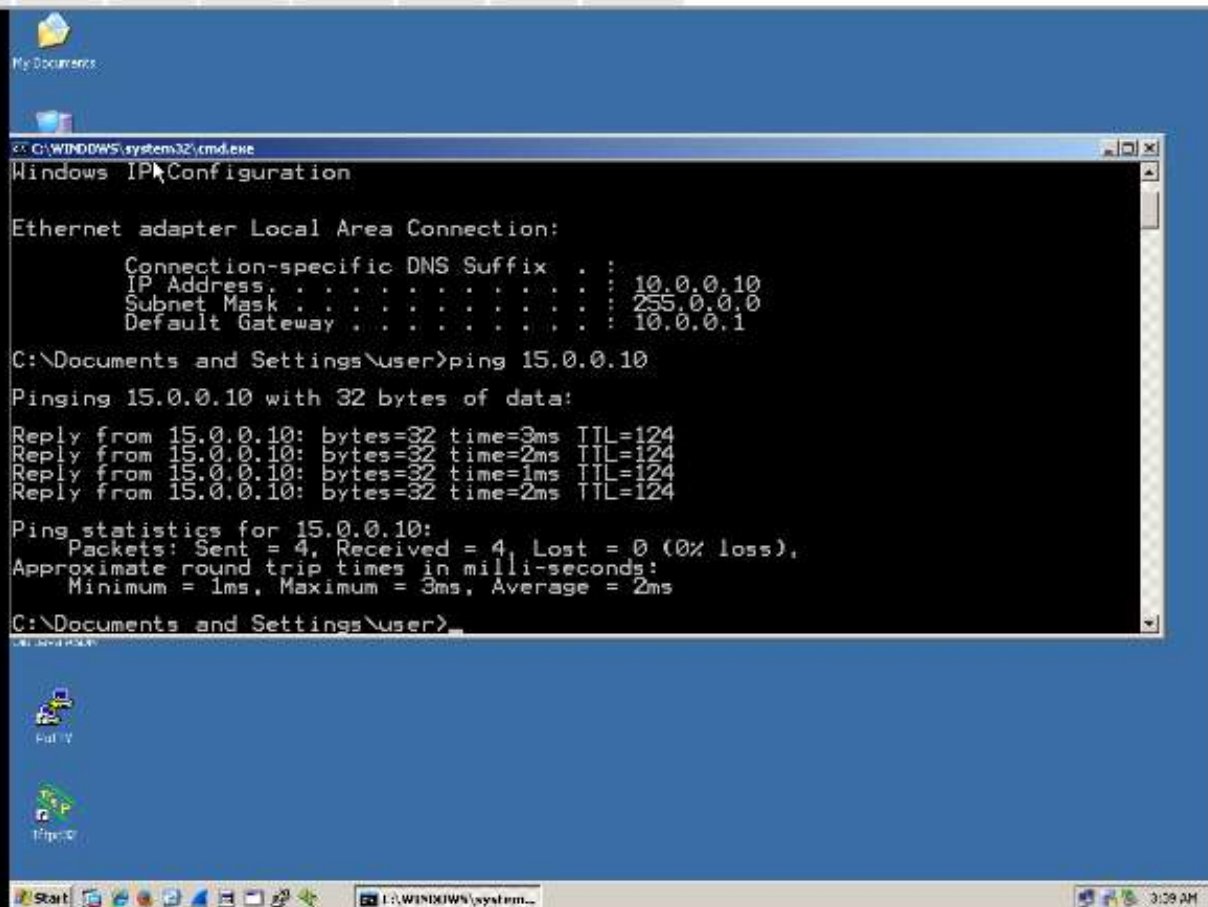




Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x



The screenshot shows a Windows XP desktop with a blue background. In the top-left corner, there is a 'My Documents' icon. The taskbar at the bottom includes the Start button, several application icons, and a system tray showing the time as 3:38 AM. A terminal window titled 'C:\WINDOWS\system32\cmd.exe' is open, displaying the following text:

```
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

terminal

Win-1 * R1 * R2 * R3 * R4 * Win-2 * R5 * R6 * Win-3 *

Press RETURN to get started.

Router>enable

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

> Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
S 15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 21.0.0.0/8 [1/0] via 20.0.0.2
S 30.0.0.0/8 [1/0] via 20.0.0.2
S 172.19.0.0/16 [1/0] via 20.0.0.2

--More-- █

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x

Router>enable

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 20.0.0.1
> 15.0.0.0/16 is subnetted, 1 subnets
S 15.0.0.0 [1/0] via 30.0.0.2
S 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
S 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2
S 172.19.0.0/16 [1/0] via 30.0.0.2
S 192.168.168.0/24 [1/0] via 30.0.0.2

Router#

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 [1/0] via 30.0.0.1
15.0.0.0/8 [1/0] via 172.19.0.2
20.0.0.0/8 [1/0] via 30.0.0.1
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 21.0.0.0/8 is directly connected, Ethernet0/2
 21.0.0.1/32 is directly connected, Ethernet0/2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 30.0.0.0/8 is directly connected, Ethernet0/0
 30.0.0.2/32 is directly connected, Ethernet0/0
162.14.0.0/16 [1/0] via 21.0.0.2
172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
 172.19.0.0/16 is directly connected, Ethernet0/1
 172.19.0.1/32 is directly connected, Ethernet0/1
192.168.168.0/24 [1/0] via 21.0.0.2

Router#

Router#

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-
2
ia - IS-IS inter area, * - candidate default, U - per-user static ro
ute
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
S    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      15.0.0.0/8 is directly connected, Ethernet0/0
>     15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
S    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.19.0.0/16 is directly connected, Ethernet0/1
L      172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
```

Router#

Router#

Router#

Router#

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM

Old Java Applet

Run TV

TIP: 300

C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
IP Address . . . . . : 15.0.0.10  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . : 15.0.0.1
```

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

```
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
```

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>

Start

C:\WINDOWS\system32\cmd.exe

3:41 AM

Terminal

Win-1 * R1 * R2 * R3 * R4 * Win-2 * R5 * R6 * Win-3 *

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM...

Old Java ASDM

PuTTY

Tripd32

C:\WINDOWS\system32\cmd.exe

```
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
```

```
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
```

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

```
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
```

```
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms
```

C:\Documents and Settings\user>

Start

C:\WINDOWS\system...

2:01 AM


```
Terminal
Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x
Router(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2
Router(config)#exit
Router#
*Aug 24 02:09:15.519: %SYS-5-CONFIG_I: Configured from console by console
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

> Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
S    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/0
L      21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
S    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.168.0/24 is directly connected, Ethernet0/1
L      192.168.168.1/32 is directly connected, Ethernet0/1
Router#
```

```
Terminal
Win-1 * R1 * R2 * R3 * R4 * Win-2 * R5 * R6 * Win-3 *
Router(config)#ip route 15.0.0.0 255.0.0.0 192.168.168.1
Router(config)#exit
Router#
*Aug 24 02:23:55.984: %SYS-5-CONFIG_I: Configured from console by console
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

> Gateway of last resort is not set

S      10.0.0.0/8 [1/0] via 192.168.168.1
S      15.0.0.0/8 [1/0] via 192.168.168.1
S      20.0.0.0/8 [1/0] via 192.168.168.1
S      21.0.0.0/8 [1/0] via 192.168.168.1
S      30.0.0.0/8 [1/0] via 192.168.168.1
       162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C         162.14.0.0/16 is directly connected, Ethernet0/0
L         162.14.0.1/32 is directly connected, Ethernet0/0
S      172.19.0.0/16 [1/0] via 192.168.168.1
       192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C         192.168.168.0/24 is directly connected, Ethernet0/1
L         192.168.168.2/32 is directly connected, Ethernet0/1
Router#
```

terminal

Win-1 * R1 * R2 * R3 * R4 * Win-2 * R5 * R6 * Win-3 *

My Documents

My Computer

C:\WINDOWS\system32\cmd.exe

Pinging 10.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

PUTTY

FTP

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

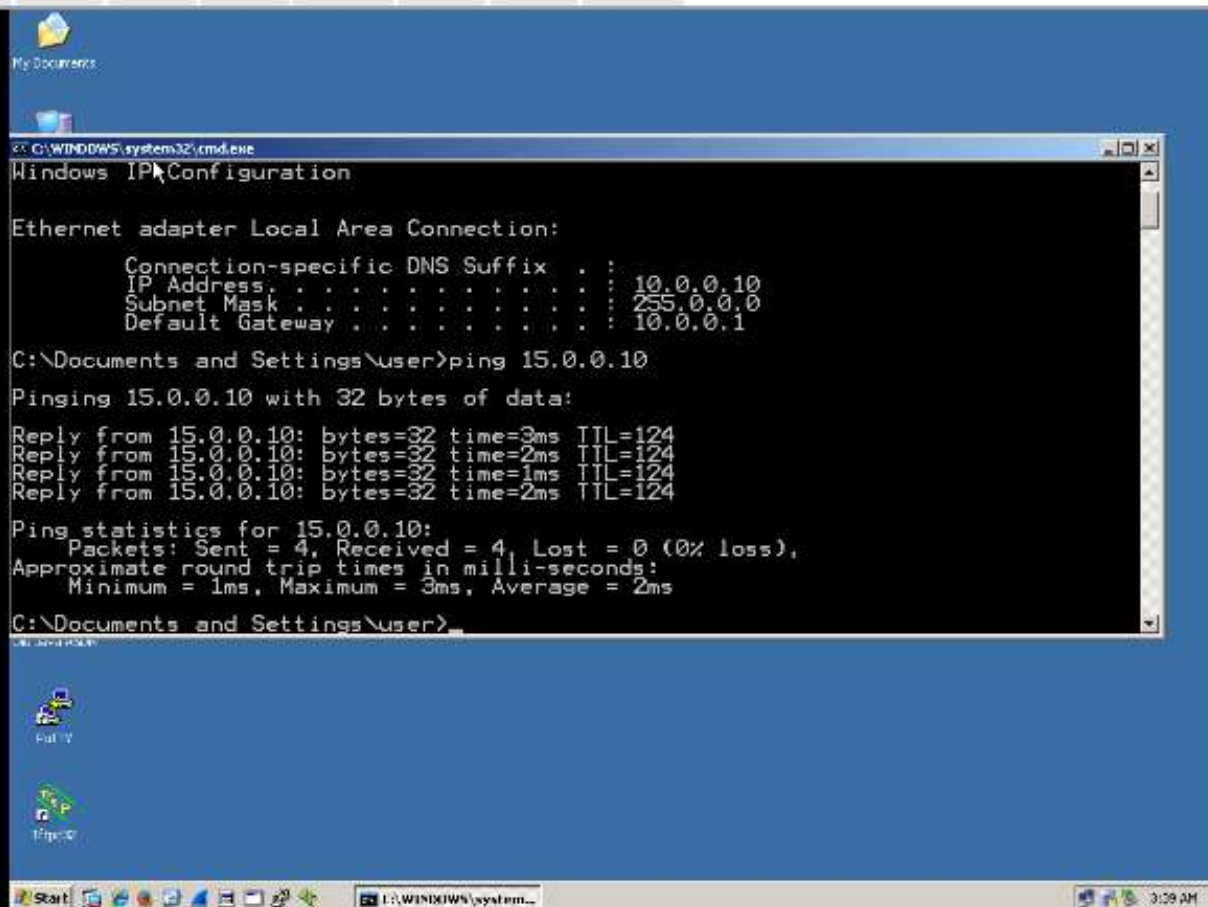
10.0.0.0/8 [1/0] via 30.0.0.1
15.0.0.0/8 [1/0] via 172.19.0.2
20.0.0.0/8 [1/0] via 30.0.0.1
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 21.0.0.0/8 is directly connected, Ethernet0/2
 21.0.0.1/32 is directly connected, Ethernet0/2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 30.0.0.0/8 is directly connected, Ethernet0/0
 30.0.0.2/32 is directly connected, Ethernet0/0
162.14.0.0/16 [1/0] via 21.0.0.2
172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
 172.19.0.0/16 is directly connected, Ethernet0/1
 172.19.0.1/32 is directly connected, Ethernet0/1
192.168.168.0/24 [1/0] via 21.0.0.2

Router#

Router#

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x



```
My Documents

C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

Start | 10:00 AM

terminal

Win-1 * R1 * R2 * R3 * R4 * Win-2 * R5 * R6 * Win-3 *

Press RETURN to get started.

Router>enable

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

> Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
S 15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 21.0.0.0/8 [1/0] via 20.0.0.2
S 30.0.0.0/8 [1/0] via 20.0.0.2
S 172.19.0.0/16 [1/0] via 20.0.0.2

--More--

Terminal

Win-1 x R1 x R2 x R3 x R4 x Win-2 x R5 x R6 x Win-3 x

Router>enable

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 20.0.0.1
> 15.0.0.0/16 is subnetted, 1 subnets
S 15.0.0.0 [1/0] via 30.0.0.2
S 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
S 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2
S 172.19.0.0/16 [1/0] via 30.0.0.2
S 192.168.168.0/24 [1/0] via 30.0.0.2

Router#